Photosynthesis

Overview

Photosynthesis is the process by which plants, algae and some bacteria turn light energy into chemical energy. The overall result is glucose and oxygen, made from carbon dioxide and water using light.

Where it happens

In plants, photosynthesis takes place in chloroplasts, found mainly in leaf cells. Chloroplasts contain chlorophyll, the green pigment that captures light. Chlorophyll absorbs mostly red and blue light and reflects green light, which is why leaves look green.

Light-dependent reactions

The first stage happens in the thylakoid membranes. Light energy is used to split water molecules. This releases oxygen as a by-product and produces two energy carriers, ATP and NADPH.

The Calvin cycle

The second stage happens in the stroma and does not need light directly. It uses the ATP and NADPH from the first stage to turn carbon dioxide into glucose. The cycle repeats to build each sugar molecule.

What affects the rate

Light intensity, carbon dioxide concentration and temperature change how fast photosynthesis runs. Very high temperatures slow it down because enzymes stop working properly.